THE LENGTH OF THE QUOTIENT BY POWERS OF AN ELEMENT AVOIDING THE MINIMAL PRIMES IS AT MOST N TIMES THE LENGTH MODULO THE ELEMENT ITSELF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author's proof span ends at source line 38013 without restoring the missing full stop of the last sentence 'Hence we get P(n) = P(n-1) + P(1) for all n >= 1'; the ending is delivered as it stands. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Finite length of a module supported at the closed point, and additivity of length in a short exact sequence. used at proof lines 37991-37993 and 38003-38004. Makes P well defined and gives the inequality in the induction step.
- Exactness on the left of the sequence R/x^{n-1}R -> R/x^nR -> R/xR -> 0 when x is a nonzerodivisor. used at proof lines 38010-38012. Upgrades the inequality to the exact recursion and hence to equality.

Source reference. Original lemma 00IV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 693–693. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00L5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 775–775. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension one, and let $x \in \mathfrak{m}$ be an element not contained in any minimal prime of R. Then*

(1) *the function $P : n \mapsto length_R(R/x^nR)$ satisfies $P(n) \leq nP(1)$ for $n \geq 0$,*
(2) *if x is a nonzerodivisor, then $P(n) = nP(1)$ for $n \geq 0$.*

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Proof. Since $\dim(R) = 1$, we have $\dim(R/x^nR) = 0$ and so $\text{length}_R(R/x^nR)$ is finite for each n (Algebra, Lemma 00L5). To show the lemma we will induct on n. Since $x^0R = R$, we have that $P(0) = \text{length}_R(R/x^0R) = \text{length}_R 0 = 0$. The statement also holds for $n = 1$. Now let $n \geq 2$ and suppose the statement holds for $n - 1$. The following sequence is exact

$$R/x^{n-1}R \xrightarrow{x} R/x^nR \to R/xR \to 0$$

where x denotes the multiplication by x map. Since length is additive (Algebra, Lemma 00IV), we have that $P(n) \leq P(n-1) + P(1)$. By induction $P(n-1) \leq (n-1)P(1)$, whence $P(n) \leq nP(1)$. This proves the induction step.

If x is a nonzerodivisor, then the displayed exact sequence above is exact on the left also. Hence we get $P(n) = P(n-1) + P(1)$ for all $n \geq 1$. □